TBA3241 Social Media Analytics

AY25/26 Semester 2

Project Proposal

Group 1

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Dataset Description

Dataset Name

Twitter Interaction Network for the US Congress

Dataset Source

Stanford Network Analysis Project (SNAP)

https://snap.stanford.edu/data/congress-twitter.html

Data Nature and Scope

This dataset is publicly available, API-collected interaction data of U.S. Congress members on Twitter. It consists of public Twitter interactions, e.g. mentions or retweets, among the Congress members and was collected by researchers for their research paper published in 2023.

The tweets were collected between 9 February 2022 and 9 June 2022 (Fink, et al., 2023), and the network contains 475 Congress members represented by nodes, and 10,222 connections represented by edges.

What the network represents

The network represents Twitter interactions among U.S. Congress members with the following representations:

- Node: individual Congress member's Twitter account
- Directed edge: connection initiated by target node, to source node
- Edge weight: transmission probability

where connection refers to retweets, quote tweets, replies, and mentions.

The weighted connection is calculated using the ratio of number of interactions to the total number of tweets issued by source node. The weightage can therefore be interpreted as the transmission probability, i.e. probability that the target node is influenced by the source node (Fink, et al., 2023).

Planned Analysis

Research Questions

1. **Community differences**
    - How can the users be categorised?

- How do the categories differ from one another?
- Is the Congress Twitter network structurally centralised around a few dominant members?
- Does the network exhibit high degree centralisation or betweenness centralisation?

We could potentially compare Democratic vs Republic, Senate vs House, etc.

2. **Focus on influence**
   - Who are the most influential users, and why?
   - Is there an imbalance of influence where influence is not reciprocated?
   - Are the influential users balanced across parties?

   We can use statistics we learnt in class to examine the centrality and network structure.

3. **Cross-party influence**
   - To what extent are there cross-party influence?
   - Are there any critical bridging nodes that act as articulation point?
   - Is there a stronger influence within the same party?

   We could potentially evaluate within-group influence vs cross-party influence.

"What-if" Questions

1. **Removal of nodes**
   - How will the overall network cohesion (e.g. number of components, diameter, fragmentation) change if the most influential nodes are removed?
   - What if the bridging nodes are removed? Does the network rely on a small subset of nodes to maintain connectivity?
2. **Changes to edge weight**
   - What if the transmission probabilities are increased or decreased in different scenarios? How will this change the graph analysis and the viral centrality of the nodes?

Potential Implications

We can potentially extend our analysis to better understand the larger influence structure of the Congress in reality, and apply the same questions in political analyses, such as finding out who exerts most influence, the cluster strength, and structural fragility. Political parties could also potentially use the findings to strengthen intra- and inter-party influence, as well as optimising strategies during election campaigns.

References

Fink, C., Fullin, K., Gutierrez, G., Omodt, N., Zinnecker, S., Sprint, G., & McCulloch, S. (2023). A centrality measure for quantifying spread on weighted, directed networks. *Physica A*.

Stanford University. (n.d.). *SNAP: Network datasets: Twitter Interaction Network for the US Congress*. Retrieved from SNAP: https://snap.stanford.edu/data/congress-twitter.html

AI Tool used	**How is the tool used in the work**
ChatGPT	Prompt: “Referring to the paper "A centrality measure for quantifying spread on weighted directed networks" pdf, what does transmission probability mean in this context?” This prompt allowed us to understand the meaning behind the weightage of each edge.
ChatGPT	Prompt: “What are some questions we can ask ourselves regarding the influence between congress members? Can we show the effects of echo chamber here?” This prompt helped us consider the different ways we can approach / explore the topic of social media influence.
ChatGPT	Fixing of English grammar and vocabulary.